

0.0.1 Interacting Multipoles of Rank 3 and 3

[illegible]

Simplify deltas:

[illegible]

[illegible]

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot -\frac{1}{225} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\delta\epsilon\zeta}^A \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\delta\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\delta\delta\epsilon}^A \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\beta\delta\epsilon}^A \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& -2835 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\delta\epsilon}^A \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\gamma}^B \cdot \Omega_{\beta\delta\epsilon}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\beta\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{\gamma\gamma\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\beta\gamma\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{\gamma\gamma\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\gamma\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\gamma}^B \cdot \Omega_{\beta\beta\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\beta\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{\beta\gamma\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{\alpha\gamma\delta}^A \\
& +18 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& +9 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& +18 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\beta\gamma}^A
\end{aligned} \right] \quad (2d)
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& -\frac{1}{225} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Omega_{\alpha\beta\gamma}^B \Omega_{\delta\epsilon\zeta}^A = R_z^{-13} \cdot -\frac{1}{225} \cdot \left[\begin{aligned}
& +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^B \cdot \Omega_{zzzz}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^B \cdot \Omega_{z\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^B \cdot \Omega_{z\delta\delta}^A \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\gamma}^B \cdot \Omega_{zz\gamma}^A \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\beta}^B \cdot \Omega_{zz\beta}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\beta\beta}^B \cdot \Omega_{zzz}^A \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^B \cdot \Omega_{zz\alpha}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Omega_{zzz}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Omega_{zzz}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zz\gamma}^B \cdot \Omega_{\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zz\beta}^B \cdot \Omega_{\beta\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\beta\beta}^B \cdot \Omega_{z\gamma\gamma}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\beta\gamma}^B \cdot \Omega_{z\beta\gamma}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^B \cdot \Omega_{\alpha\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Omega_{z\gamma\gamma}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\gamma}^B \cdot \Omega_{z\alpha\gamma}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Omega_{z\beta\beta}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\beta}^B \cdot \Omega_{z\alpha\beta}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{zz\beta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{zz\alpha}^A \\
& +18 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& +9 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& +18 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Omega_{\alpha\beta\gamma}^A
\end{aligned} \right] \quad (3a)
\end{aligned}$$

Multiply Out Sum:

$$-\frac{1}{225} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Omega_{\alpha\beta\gamma}^B \Omega_{\delta\epsilon\zeta}^A = R_z^{-13} \cdot -\frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A + 1890 \cdot R_z^6 \cdot \Omega_{xxz}^B \cdot \Omega_{xxz}^A \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{xyz}^B \cdot \Omega_{xyz}^A + 1890 \cdot R_z^6 \cdot \Omega_{xzz}^B \cdot \Omega_{xzz}^A \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yyz}^B \cdot \Omega_{yyz}^A + 1890 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A + 1890 \cdot R_z^6 \cdot \Omega_{xzz}^B \cdot \Omega_{xzz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxx}^B \cdot \Omega_{xxx}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxz}^B \cdot \Omega_{xxz}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xyy}^B \cdot \Omega_{xyy}^A + 18 \cdot R_z^6 \cdot \Omega_{xyz}^B \cdot \Omega_{xyz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxz}^B \cdot \Omega_{xxz}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xzz}^B \cdot \Omega_{xzz}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xyy}^B \cdot \Omega_{xyy}^A + 18 \cdot R_z^6 \cdot \Omega_{yyy}^B \cdot \Omega_{yyy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{yyz}^B \cdot \Omega_{yyz}^A + 18 \cdot R_z^6 \cdot \Omega_{xyz}^B \cdot \Omega_{xyz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{yyz}^B \cdot \Omega_{yyz}^A + 18 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxx}^B \cdot \Omega_{xxx}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxz}^B \cdot \Omega_{xxz}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xyy}^B \cdot \Omega_{xyy}^A + 18 \cdot R_z^6 \cdot \Omega_{yyz}^B \cdot \Omega_{yyz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xzz}^B \cdot \Omega_{xzz}^A + 18 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{zzz}^B \cdot \Omega_{zzz}^A \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot -\frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A + 1890 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A + 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A + 18 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A + 18 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \end{aligned} \right] \quad (6b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot -\frac{1}{225} \cdot \left[\begin{aligned} &-4671 \cdot R_z^6 \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &+ 54 \cdot R_z^6 \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &+ 18 \cdot R_z^6 \cdot \Omega_{yyy}^B \cdot \Omega_{yyy}^A \end{aligned} \right] \quad (6c)$$

Combining everything:

$$-\frac{1}{225} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Omega_{\alpha\beta\gamma}^B \Omega_{\delta\epsilon\zeta}^A = \left[\begin{aligned} &+\frac{519}{25} \cdot R_z^{-7} \cdot \Omega_{yzz}^B \cdot \Omega_{yzz}^A \\ &-\frac{6}{25} \cdot R_z^{-7} \cdot \Omega_{xxy}^B \cdot \Omega_{xxy}^A \\ &-\frac{2}{25} \cdot R_z^{-7} \cdot \Omega_{yyy}^B \cdot \Omega_{yyy}^A \end{aligned} \right] \quad (7)$$